Manual

Master Data / Gage Management

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Contents

1 Master Data / Gage Management

Purpose

This component supports creating gages (resources of type PRM) or gage groups (resource families).

Implementation Considerations

This component should be used to document which gages or gage groups are used to execute the inspections. Additionally, it forms the basis for automatic measurement value collection through interface equipment.

Integration

This component serves the components:

- Inspection planning for production control inspections
- Inspection planning for goods receipt inspection
- Inspection planning for initial sample inspection
- Measurement data interface for quality data

Features

The following functions are available:

- Maintenance function to create and modify relevant master data (gage groups or gage families, storage locations etc.)
- Maintenance function to create and modify gages and assign these to status, cost center, resource family, owner, inventory number, engraving number etc.)

2 Configuration of Workplaces and Resources

Summary

Menu	Master data → Resources → Resource configuration Master data → Workplaces/ Machines → Workplace configuration
Transaction code	res
Function authorization	mdres

The resource configuration is the central function for resource management in MES.

Usage

The master data for both workplaces and machines as well as for other resources (tools, DNC resources, etc.) are managed here. The resources are roughly classified according to resource type. This type is also connected with corresponding functions and applications that open other functional components of the MES especially for the respective type.

Integration

This application can be used to view the resource information of all of the resource types present in HYDRA. However, the data record maintenance depends on the resource type. In this way, depending on the resource type, not all fields can be maintained and not all resources can be created or deleted.

Based on the resource type, other applications are present in the MES that are specially customized for these types. For example, the application package of machine data collection is based on resources of type "Machine".

In addition to the resource configuration, the resource overview application is present, which does not permit data maintenance, but does enable administration operations for daily handling of resources such as the stock transfer of a resource.

Requirement

Before a workplace or machine is created, a year model/ shift calendar must be created. To use the various resource types in a meaningful way, the advanced licenses for these types must also be available.

Selection criteria

The following selection criteria are available in the application:

Resource from ... to ...

This selection criterion refers to the resource. Wildcards (placeholders *) can be used.

Short designation

Short designation of resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have resource type MNR, while in the context of customizing other resources can be given individual resource types. Predefined resource types include:

DNC	NC-/ DNC program
DOC	Document
ENE	Energy meter
ENT	Extraction device
ENT	Extraction device
MNR	Workplace/ Machine
PAC	Packaging, transportation container
PRM	Inspection and measuring equipment
PER	Production staff / general
PRU	Set up staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

Using the predefined resource types is recommended.

The detail resource information is adapted depending on the resource selected in the table overview.

Designation

Designation of the resource.

Group

Workplace/ Machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of resource.

Short name

Short name of the resource

Resource family

Family to which the resource is assigned.

Responsibility area

Responsibility area to which the resource is assigned.

Storage location

Master storage location of resource.

MD user fields

MD user fields 1- 6 of the resource. If a resource family is selected in the selection panel, the field names are displayed based on the assigned user field definition.

Field descriptions

This detail application includes three main tabs:

- Resource configuration
- Resource list
- Resource attributes

Main tab "resource configuration"

The configurations and master data of resources are defined here.

General tab

Resource type

Resource type of the resource. When the HYDRA system is delivered, some resource types are predefined. Further resource types can be created within the scope of HYDRA customizing.

Resource

The number of the resource or workplace to be entered is input in this field.

The maximum number of places of this number is as follows, depending on the resource type:

- Resources of type MNR: maximum of 8 places
- Resources of type <> MNR: maximum of 20 places

Permitted characters include ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890/_.-+#. Spaces and other special characters are not allowed. For technical reasons, * (asterisk) and % (percent) can be input, but are nonetheless not permitted because they are not valid characters. When the input field is exited, lower case letters are automatically transformed into CAPITAL LETTERS.

Notes regarding workplaces/ machines (resource type MNR):

For technical reasons, for resources of type MNR there is no check for the maximum number of places. For this reason it must be ensured that the length of the resource number (= workplace/ machine number) has a maximum of 8 digits.

If the option "Numeric machine number" (HYDRA basic settings) is activated for use on DOS-based terminals, it must be ensured that the resource number (= workplace/ machine number) includes only numerical digits and that the length of the number is exactly 8 places. If necessary, when creating the workplace/ machine, zeros must be added to the beginning of the number to extend it to 8 digits.

Short designation

Short designation of resource. This field can only be used with workplaces/ machines (resources of type MNR).

Designation

This field is used to assign a short, unique designation for each resource. This designation is displayed in reports and overviews and at the terminal and it is useful for orientation.

Responsibility area

Responsibility areas are used such that in various evaluations, the user is only shown the data to which he/she has access according to his/her responsibility area authorization.

The responsibility area can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.

Cost center

The cost center to which the resource belongs is entered in this field.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information that functions as a comment.

Acquisition date, acquisition costs

Additional information that functions as a comment.
The currency is configured across the system in the HYDRA basic settings.

Storage location

Location at which the resource is stored when it is not being used (home storage location).

In connection with the material and production logistics (MPL), the material buffer specified in this field is used during input batch logon for reposting the logged on input batch(es) from the previous material buffer to this material buffer (upstream of the machine).

Delivery date, start-up date, guaranty date

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

External designation, resource type designation, usage, purchase order number

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

Supplier and party in charge including detail fields

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

Workplace configuration tab

This tab is only available if a resource of the type "MNR" is selected.

Workplace master data

Workplace category

N Machine

P Workplace

Definition as machine or workplace. As regards processing, both of these categories are identical if only BDE or MDE and PDV are in use.

J Machining center

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined purely as a CAQ inspection station without affecting the BDE or MDE statistics.

R Reel-based manufacturing (MPL-RF only)

S Cutting unit (MPL-RF only)

The categories "Reel-based manufacturing" and "Cutting unit" and their functions are described in detail in the MPL-RF product documentation.

D Parallel output batches (MPL only, starting with MPL 7.2)

Parallel output batches can be produced on the machine for an operation with a batch management requirement.

C Packing station (MPL-PAL only, starting with MPL 7.2)

Specific posting functions are used on the machine for mapping a packing station. The functions are described in detail in the MPL-PAL product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

Workplace type

E Single workplace

G Group workplace

Please note

Terminals can be assigned to both group and single workplaces. However, in this case it must be noted that the terminal is set to operation mode "BDE" or the option **Processing** is set to "BDE processing" when assigning workplaces to terminals.

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this identifier is set, the machine/ workplace has been (logically) deleted. The following modifications are no longer permitted in this case:

- Order postings on the terminal
- Order posting on the console/MOC (e.g. using the Order overview function)
- Modifications in event maintenance

Furthermore, the machine/ workplace is no longer displayed in the *Machine overview* or in the graphic planning board of the HYDRA shop floor scheduling module (HLS).

Company

This field is used to differentiate the individual machines/ workplaces. Within the system, it also functions as report/evaluation option to some extent.

Group

Assignment of the workplace/ machine to a logical group. In the context of the planning this has to do with a capacity group in which the primary capacities are summarized.

If a new workplace is created, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If there is no capacity group yet, then it is automatically created and then the workplace is automatically assigned to it.

Category

The category of the machine is entered here. From this category, a validation check can be activated according to the customizing configuration BDE: Master data > Order configuration > Order types, Plausibilities tab, option "Check for specifications in backlog of orders" *(*value *Category)*.

Year model

A valid year model must be entered here. During entry, times to be posted are compared with this shift model. If no planned year model is stored in the HLS tab, this shift model is also used in the HYDRA shop floor scheduling module (HLS).

Standard rate machine

The arithmetical standard rate of machines can be entered here for calculations. In the HYDRA shop floor scheduling module (HLS) this value is used for some (evaluated) key figures.

Standard labor rate

The arithmetical standard labor rate can be entered here for calculations. In the HYDRA shop floor scheduling (HLS) this value is used for the key figure "Evaluated labor utilization".

Performance level

The performance level of the workplace/ machine can be entered in percent in this field. This value is considered in the HYDRA shop floor planning and in the evaluation of material requirements when calculating the remaining run time.

Incentive wages indicator

Defines the type of calculating incentive wages. Mostly, this flag is used together with the incentive wages based on formulas for customer-specific configurations. In addition, the "incentive wage indicator" can also be used as selection criterion for the determination of wage types within the incentive wage determination.

This field should be left empty if the incentive wages determination is not in use.

The incentive wages indicator G=group piecework has a special meaning. If the workplace/machine has this flag, a premium group needs to be assigned every time an order is logged on . This can be performed either by the "assignment of premium groups" option of the incentive wages determination or, optionally, by an additional field in the terminal dialog for logging orders on. If no assignment is available, the logon of the order will be rejected by issuing a validation error.

Therefore, the incentive wage indicator G = Group Piecework may only be assigned if the group premium conditions are met in the incentive wages determination, as otherwise orders can no longer be logged on!

The meaning of the other incentive wages indicators is specified according to the customer's requirements while customizing the system.

File

It is possible to assign a graphic to every machine/ workplace. Among other uses, the graphic is displayed in the workplace overview or in the AIP. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.

In the path configuration, the path must be configured using the PATH identification "MOCWPIMG"; the length of the file name for graphics files is limited to 12 places (8.3 notation). Note for Unix installations: Please use lower case letters only for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, the maximum capacity in KG can be defined here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if the license for creating gages has been purchased.

Entry

Display 3rd list

A third list can be displayed/ activated in the basic screen on a Windows-based terminal (CTWIN / AIP) using the options displayed here. Depending on the options set, users can switch between the respective lists on the terminal. The following settings are possible. Note here that the display in the lists depends on the module set:

→ Input material (MPL): Logged on input materials/ batches are displayed.

→ Resources (WRM): Logged on resources and tools are displayed.

→ Staff (BDE): Logged on staff is displayed.

→Output material (MPL): Produced output batches are displayed.

Show material/ PRT list when OP is logged on

This option is only relevant in connection with WRM and resources logged on at Windows-based terminals (CTWIN / AIP).

If this option is set, when an OP is logged on, a specific logon dialog is called. This dialog contains a component/ PRT list in which resources are displayed that meet at least one of the following conditions:
- the identifier "Logon on terminal" is set on the resource type;
- the identifier "Log on with OP" is set to "Explicit logon"
- the resource is a so-called "required resource" (identifier on the resource).

Please note: As long as the workplace is relevant for MPL, material components are also displayed in this list.

Sequencing list

This option is used to define which operations are to be displayed in the sequencing list on the terminal. The following settings are possible:

S Basic setting. The value is taken from the option of the same name in the *HYDRA basic settings*.

M Pool of workplaces. Only the operations planned at the workplace are displayed in the sequencing list on the terminal.

G Pool of workplaces and groups. The OPs displayed in the sequencing list on the terminal are those that are either planned at the current workplace or another workplace in the group or are still located in the pool of groups.

K Pool of workplaces and categories. Only those operations that are planned at workplaces of the category are displayed in the sequencing list on the terminal.

H Group control. The OPs displayed in the sequencing list on the terminal are those that are either planned at the current workplace or another workplace in the group.

Number of OPs in sequencing list

The maximum number of OPs that are to be displayed in the sequencing list on the terminal can be stored here.

Compulsory sequence

This option can be used to specify whether or not logging the OPs on in the planned sequence is compulsory. The following parameters are permitted:

N Disabled

J Enabled/active

If the parameter is enabled when the OP is logged on a check is made as to whether or not there is an OP in the order pool for this machine/ workplace that is planned for the same time or previous to this OP in the sequence, but was not yet started (i.e. status = V/prepared). If yes, the OP logon will be rejected.

Please note: If the planning in HYDRA is done using order sequencing (menu ADE: Planning > Order sequencing), a configuration of the sequencing list that is not equal to "M" (pool of workplaces) and an active compulsory sequence can result in a combination that does not make sense.

Dialog control

To meet this requirement, a dialog control that deviates from the standard behavior can be defined at the workplace in the dynamic dialog configuration on the Windows-based terminal (CTWIN / AIP) and then a corresponding reference can be made to it in the dialog.

This configuration is only relevant and can only be used in the context of the HYDRA customizing. Otherwise it has no significance.

Logon of several OPs

If several different operations are to be processed on the machine, this identifier must be selected. Otherwise HYDRA only allows one order/ operation to be logged onto the machine.

Possible values:

J As many OPs as desired can be logged on at the same time.

Please note: On a machine to which a terminal with operation mode MDE is assigned, a maximum of 20 operations can be logged on at the same time. If more than 20 operations must be logged on at the same time, this limitation can be removed after a review by MPDV.

N Only 1 OP can be logged on

1...9 A maximum of n OPs can be logged on

Posting

Quantity posting to person

If selected, this function is used to post the quantity of order interruptions/ logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to persons can be found here.

Posting on OPs that are not logged on

If selected, a partial confirmation/upload, interruption or logoff for an OP can be performed on this machine, even without a previous logon.

Posting the machine time in connection with operations logged on at the same time

If set, the machine time for OPs logged on at the same time is posted as a proportion on the operations and persons:

J Proportionate posting on OP and person acc. to the number of OPs

N No proportionate posting. If the option is not set, every operation receives the complete machine time.

V According to the default quantity of the OPs. Here it must be ensured that the default quantity (target quantities in primary quantity unit) in the operation > 0.

Z According to the standard time of the OPs (available starting with ADE 7.3). Here it must be ensured that the standard time (processing time) in the operation > 0.

Please note:

This option is also evaluated for group workplaces; in general, the option should not be set for them.

Automatic logoff of personnel when shift ends

This identifier is only relevant if an "X" is set for the identifier of the same name on the order type. The configuration is carried out by MPDV while customizing the system. Otherwise it has no significance.

This identifier is used for the more detailed configuration of the personal data collection at MDE workplaces. Because fully automatic shift ends are generated by the terminals when using the HYDRA MDE, a setting can be made here regarding whether the people logged onto the workplace are automatically logged off at the end of the shift or if they remain logged on.

J Always log off staff when shift ends

N Always save staff when shift ends except for manual logoffs

X Evaluate the person's settings. Now the person will be searched for the corresponding setting

Automatic OP posting when shift ends

This configuration is only relevant in the context of the HYDRA customizing by MPDV and can only be used then. Otherwise it has no significance!

This identifier is only relevant if an "X" is set for the identifier of the same name on the order type.

J Interrupt and log on again at beginning of shift

N Interrupt

MPL

Further information about the HYDRA module MPL can be found in the corresponding MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the posting dialogs on the terminal.

Possible values are:

N No batch processing

L Batch tracing (input/ output batches) in the context of HYDRA MPL/MPL-RF

D Throughput batch processing in the context of HYDRA MPL/MPL-RF

J BDE batch management

The following functions are only available in connection with the Material and production logistics module and are supported only on the Windows-based terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

→Subsequent material buffer

If a material buffer is specified in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log operation off) is automatically populated with this value.

If no material buffer was entered in the entry dialog (e.g. deleted from the input field), the output batch is automatically posted to this material buffer.

Automatic generation of batch number

If this identifier is set, a batch number is automatically generated for the output batch to be produced. Otherwise during the operation logon or output batch change, the input of the batch number of the new output batch to be produced is expected.

Please note: If, in the field *Batch management* the setting D (= *Throughput batch recording*) is activated, the value for *Automatic generation of batch number* is automatically set to "J". In this case, manual entry of the batch number is not possible.

Consumption balance

During the OP logoff, an additional dialog (V_BLZ) is opened displaying the material components and their consumption quantities in relation to the current OP logon. In this dialog, the operator has the option to log off input batches that are still running. The option is only enabled, once the consumption balance has been activated for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport is started from the material buffer in which the output batch is produced. The option set here is overridden by the configuration of the relevant option within the material type.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when an operation is planned for a machine using the shop floor scheduling module. Transportation is started from the output material buffer of the preceding operation. The option set here is overridden by the configuration of the relevant option within the material type.

HLS

Further information about the HYDRA module HLS can be found in the corresponding HLS documentation.

Planning function

This identifier specifies whether or not a workplace or a machine will be displayed and if so, in which MOC planning function.

P	Planning in the graphic planning board of the HYDRA shop floor scheduling (HLS) or in the graphic order sequencing (GAV), i.e. the workplace is planned using the HLS or the graphic order sequencing and for this reason it is visible there, but not in the order sequencing table (AVG). Please note: Whether or not a workplace is displayed in the HLS or the graphic order sequencing depends on other settings as well: - Assignment to a group identified as a "capacity group" - Responsibility area authorization for this workplace - Planning profile

H Only relevant when using the HYDRA shop floor scheduling module (HLS).

Same as P. Furthermore, the workplace is also considered in the automatic (server-based) planning. The activation of the automatic (server-based) planning can only be performed while customizing HYDRA.

T The workplace is only considered in automatic, server-based planning; in the graphic planning board the workplace is not displayed. The activation of the automatic (server-based) planning can only be performed while customizing HYDRA.

A Planning in the order sequencing table (AVG), i.e. the workplace is planned using the AVG module.

N No planning; the workplace is displayed in neither the order sequencing AVG table nor in the graphic order sequencing nor in the HLS module.

Planned year model

A special year model used only for planning in the HYDRA shop floor planning (HLS) can be entered here. It does not affect entry and posting in the ADE/MDE module. If no planned year model is defined, then the BDE year model (*Master data* tab) is used for the planning.

Availability

The available capacity of a workplace/machine can be defined here. The default value for the available capacity is 1000 [per mill].

In the HYDRA shop floor scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. only one order/ operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs the user about the double assignment; automatic assignment always assumes a single assignment.

This setting can be used to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, the available capacity should be set to 2000 [per mill] in this field.

If nothing is entered in this field or if the value 0 is entered, the system interprets this as the standard value of 1000 [per mill].

Utilization of this function is subject to the corresponding license and requires further customizing services by MPDV, depending on the relevant field of application.

Quantities tab

This tab is only available if a resource of the type "MNR" is selected.

Conversion factors to basic quantity

At the machine or workplace, the quantities can be collected in various quantity types and accounts. In general, the following quantity accounts are supported:

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported for each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The actual use of several quantity types or accounts depends on the system design. For example, for manual entry of rework quantity, a corresponding input field must be configured in the entry dialog (by customizing).

Automatic quantities can only be entered in the "primary quantity" unit.

Quantity units and conversion factors for base quantity

A quantity unit is defined for each quantity type. The alternative quantity accounts can be entered directly (manually). If this is the case, automatic conversion is not carried out.

If the alternative quantity accounts are not entered manually, a conversion into the alternative accounts is carried out on the server based on the conversion factors or the units configured in the MOC machine master data.

Basis for HYDRA-MDE quantity conversion

The basis that will be used for quantity conversion is set here.

A Use the conversion factors of the OP that is logged on. If no operation is logged on, the quantity conversion from the machine/workplace configuration is used.

M Use conversion factors from the workplace configuration for quantity conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit in which the entry at this machine/ workplace is primarily carried out. In case of automatic entry of quantities, these quantities are generally viewed as primary quantities.

If an automatic quantity conversion into another quantity type is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Here indicate the secondary quantity unit in which quantities are to be posted to the workplace/ machine. If an automatic quantity conversion is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Here indicate the tertiary quantity unit in which quantities are to be posted to the workplace/ machine. If an automatic quantity conversion is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Here indicate the base quantity unit in which quantities are to be posted to the workplace/ machine.

Manual entry of quantities, yield

Manual entry of yield

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows-based terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of yield as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, scrap

Manual entry of scrap

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of scrap as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, rework

Manual entry of rework quantity

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of the rework quantity as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, open quantity

Manual entry of open quantity

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of open quantity as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

MDE configuration tab

This tab is only available if a resource of the type "MNR" is selected.

Monitoring

Monitoring type

The following types of monitoring can be selected:

Monitoring by operating signal

No monitoring

Cyclical monitoring

If **cyclical** or **operating signal** monitoring was selected, a malfunction can only be entered if a request is made using the terminal ("Assign malfunction"). If **no automatic monitoring** is specified, a new machine status can be specified at any time.

In **cyclical** monitoring, when a counting pulse occurs, an automatic switch is made into the "Production" status. If operating signal was selected and when the operating signal is set, a switch is made into **Production**. If no automatic monitoring was selected, the "Production" status must be assigned manually.

→Entry of malfunction reason required with specified delay time in [s]

(Only with terminal type CT 73x, CT 83x, not with master terminal/DS-100)

If a downtime without a reason was recognized, after the specified delay time the input window for "Change machine status" opens automatically on the terminal. If the terminal goes back into production, the window still remains open.

If a machine status is input now (during production), this input activates a reposting event that reposts the most recently recorded status from "General disturbance" to the new status that was input. If the reposting is correct, the window closes; otherwise, it remains open.

However, if a downtime is recognized again (with or without a reason), the previously noted status can no longer be reposted. The window closes automatically.

If another downtime without a reason is recognized and the delay time has expired, then the input window opens as described above.

If a downtime without a reason is recognized and the machine switches into production before the delay time expires, then the automatic request for a malfunction reason is not carried out.

Important note:

This reposting only affects the HYDRA machine data collection; online correction of the resource performance accounts of the currently running OP is not possible !

Note regarding data maintenance:

All machine status modifications are displayed in the tabular event maintenance of MOC. However, the reposting event is locked and cannot be edited.

In order to perform recalculations correctly now with respect to orders and machines, the original event with the status "NOT ASSIGNED" must be changed to the correct status.

The reposting event does not affect the recalculation!

Minimum malfunction duration

If the **operating signal** monitoring type was selected, the duration that a malfunction must last until it is recognized and logged in as a malfunction is specified in seconds in this field.

Minimum cycle time

If **cyclical** monitoring was selected, a minimum cycle time can be specified in seconds in this field.

From this minimum cycle time and the target cycle stored in the (logged in) operation and compensated with the cycle extension, the terminal determines the maximum value and uses it as the cycle time specification.

If both the minimum cycle time and the target cycle stored in the operation are 0, the cycle time specification is set to 60000 seconds [per 1000 machine cycles].

Cycle extension

If **cyclical** monitoring was selected, the percentage for extending the target cycle time must be input here in a range from 0 to 5000.

The target cycle stored in the (logged in) operation is compensated with this percentage. In this way, a value less than 100 indicates a shortened cycle; a value greater than 100 indicates an extended cycle.

Number of target cycles

If **cyclical** monitoring was selected, here the number of cycles (0 to a maximum of 9) can be indicated according to which the terminal automatically switches from a status not equal to production into production status <u>within the cycle time</u> (prerequisite: no production lock is currently set in the status not equal to production).

With some production processes, there are machine cycles in set up phases. By setting a value greater than 0, the current machine status does not change immediately. Please note: Until a switch is made into production, the quantities entered in this case are neither posted to the yield nor the scrap quantities.

Cycles to be evaluated

This entry should be populated with 0.

Note regarding data maintenance:

All machine status modifications are displayed in the event maintenance dialog. However, the reposting event is locked and cannot be edited.
In order to perform recalculations correctly now with respect to orders and machines, the original event with the status "NOT ASSIGNED" must be changed to the correct status.
The reposting event does not affect the recalculation!

Administration

Posting during prod. lock

This setting can be used to set the type of posting of the counting pulses collected during the so-called production lock. This configuration takes effect with all counters configured as "Yield".

Posting as scrap

If the counting pulses were configured on the counter, they are offset against the partitioning/ pulse factor and posted as scrap. However, any defined allocations with another quantity account are not carried out in this case.

Posting as yield parts

Posting of the counting pulses as yield

No posting

No quantities are posted during the production lock.

→→→Pulse factor specific to machines

The pulse factor is used, for example, if lengths (e.g. from a wheel) are to be collected.

For machines for which a discrete or integral number of quantities (e.g. pieces) are entered per pulse, the value should always be set to 0. In this case, the pulse factor is not used for the evaluation. That means that the posting of the number of cycles corresponds with the actual pulses transferred via the MSS (machine interface).

The signals transferred from the machine (counting pulses) are collected by the MSS. The quantity is calculated and posted as follows according to the configured number of pulses:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be evaluated as a fraction. In this way, in the quantity calculation the pulse is included as a denominator while the partitioning is considered a numerator.

Pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock* > scrap) are interpreted as scrap. The scrap quantities are also calculated using the formula given above.

Partitioning specific to machines

The partitioning specific to the machine is indicated here. This is included in the quantity calculation in a multiplication with the partitioning entered in the operation. If this is not desired, the value 1 must be input here.

Extended weekend automatic

If this option is selected and with the corresponding configuration the status that was available before status 999 was activated will be assigned at shift start.

Please note:

To set the option, the workplace must already be assigned to a terminal.

Detailed information about the automatic activation of status 999 can be found in the chapter *Day types* .

Wait. period short-term disturb.

To improve the overview, e.g. in machine history, a short-term malfunction status can be configured per machine/ workplace. In the context of machine monitoring this status serves as a "container" for any unconfirmed statuses that existed only for a certain (short) period of time.

If a downtime is automatically recognized on the terminal and the machine automatically goes back into production, a check is made regarding whether or not this disturbance is shorter than the time period configured here for short-term malfunctions.

If this is the case, the malfunction, which does not yet have a reason, is given the status (reason) configured on the machine as the status for "short-term malfunctions".

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

The terminal outputs used for production statuses are indicated in these fields

Machine lock output	is set if the status "not assigned" occurs or a status in which the option "machine lock" is set.
Target quantity reached output	is set if the target quantity of the OP is reached.
Machine downtime output	is set if a status not equal to P**roduction** was recognized for the machine**.** When changing into production, the output is immediately reset to 0.
Free I/O	Free input/ output for customer-specific customization.

These statuses can be used for connecting a monitoring light or a horn, for example.

The assignment of an output by entering the corresponding number in one of the fields specifies which relay is interconnected by the terminal when the predefined status occurs. If "0" is entered, no action occurs. Note that an output on a terminal may not be assigned more than once.

Please note

- The specification regarding the status in which the machine lock is to be set is carried out in the context of the **Status assignment**.

- When the machine lock is activated using the available relay output of a DS 100, in general the value "1" is to be entered in the input field. In this case, the machine lock is set if a correspondingly configured status occurs <u>and</u> in the status <u>not assigned</u>.

Output batch change

Customer-specific assignment of an input with an automatic output batch change (MPL). As a default, the field should be left at 0.

PDE (Process Data Collection)

Collect process data

This parameter specifies whether or not process data are recorded for this machine. Process data cannot be recorded for this machine if this parameter is not set at a machine.

External connection

External connection

If this machine is assigned to a master terminal, the following selection of connection options is available:

Arburg control system	Arburg connection (only available if HYD-ALS is licensed)
Engel interfacing	Engel machine connection
No external device	No connection of an external device
DS100	DS100 connection
MT3	MT3 connection
PDE	Process data collection

If a DS100 or MT3 connection was activated, the device address field can be selected. If the option Engel interfacing is selected, the serial number field can be selected. The option Arburg control system enables the class field.

Note regarding the combination of connections on a master terminal:
"DS 100" and "No external device" allowed
"MT 3" and "No external device" allowed
"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)

The serial number of the connected Engel machine is entered here. A prerequisite for using Engel machines is the setting "EMS machine interface" in the HYDRA basic settings..

Device address

This field can be selected if a DS100 or MT3 connection was selected. The device address of the sub-bus participant is entered here.

Resource configuration tab

Resource master data

Type

Identifier regarding the type of resource:

Resource: A resource can be uniquely identified but is also actually present. It always has the number 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then the value in the field *Number* can be changed from 1 to another positive integer value. Anonymous resources cannot be posted because the actual reference to precisely one resource is not possible. Please note the information in the chapter *Anonymous resources*.

Required resource: A required resource is a substitute for one or more actual resources that can be identified. The resources that a required resource represents are specified in the configuration *WRM: Master data* > *Required resources*. The number results from the number of actual resources assigned to it.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future use.

Version

Modification number; the program version can be stored here for resources of type DNC.

Number

This field can only be edited if it contains an anonymous resource and the identifier *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is automatically calculated for required resources.

Resource family

Assignment of a resource family. If the resource family is subsequently changed, an information dialog appears as a warning because user fields can possibly be assigned with the resource family.

Target utilization

Cycles

The target cycles serve as additional information regarding how long the resource is to be used.

Runtime

The target runtime serves as additional information regarding how long the resource is to be used.

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection.

Original partitioning

Partitioning of the tool (= multiplicity or number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This partitioning can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with a cycle due to a tool defect.

The posting of cycles to the tool is always carried out based on the current partitioning.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection.

Partitioning due to cavities

Setting the flag "partitioning due to cavities" causes the system to (re-)calculate the fields "current partitioning" and "original partitioning" using the assignments in cavity management. Then the fields can no longer be changed manually.

Log on with OP

This identifier is used to control whether or not the resource is logged on if it is assigned as a component in the list of production resources and tools for the operation. Possible values are:

None: The resource is not logged on.

Implicit: The resource assigned to the operation as a production resource and tool is automatically (implicitly) logged on with the operation; an explicit logon or change in logon status is not possible.

Explicit: The resource assigned to the operation as a production resource and tool can be logged on explicitly or another resource can be logged on instead. If the resource is not explicitly logged on or if no other resource is explicitly logged on, the current resource is implicitly logged on; in this way, the current resource serves as a "default".

Please note:

If another resource is explicitly logged on, it will be logged on for the resource that has the same **resource type** in the list of production resources and tools of the operation. For this reason, an explicit resource logon is only possible for resources that are included in the list of production resources and tools of the operation as a requirement. In this way, no resource can be logged on for which there is no requirement (= entry) in the list of production resources and tools.

In general, this identifier should be inactive for resource type DNC because a specific processing exists for it in the HYDRA module DNC (NC programs are logged on separately).

Resources that are defined in the BOM of the machine are also logged on.

Parallel logon/ planning possible

The tool can be logged on/ planned in parallel.

Caution: A resource can be logged on multiple times to only one machine. Consequently, the identifier "Parallel logon possible" refers to several different OPs on one machine.

Post to resource

Indicates whether or not the quantities and times are to be posted to the resource. Due to a high degree of complexity, this identifier should only be provided for those resources that will actually be evaluated later.

Planning

Setup time

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Retooling time (teardown)

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

No processing. The configuration option of the same name stored in the resource type is used for taking the resource allocation in the HYDRA shop floor planning into consideration.

Evaluation

Consider in evaluations

Reserved for future use.

File

File exists

Shows whether or not the file is stored in the specified path. The files are checked by a cyclic process and the corresponding flags are set subject to whether or not the file is available.

File name

File name; without file extension for DNC. The file extension is added based on the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Two comparison resources may be indicated here for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resources to be compared

Resource type 1

Resource type of the resources to be compared

Resource 2

Resource number of the resources to be compared

Resource type 2

Resource type of the resources to be compared

Accuracy

More detailed information on measuring accuracy and measuring range may be entered here for test equipment resources.

User fields tab

User fields offer the possibility to store further customer-specific information to MES besides the available fields in MOC standard. The tab provides eight sub tabs each of which providing eight user fields. The so called user field key determines, which user fields are involved and which meaning they have.

Object type

User fields are configured for the relevant resource type, e.g. MNR for the workplace/machine.

User field key

Every user field key describes a combination of user fields. How the user field key is managed (and thus the meaning of the fields) varies for the individual objects. User field keys are defined together with the customer while customizing the system.

User fields

The following user fields are available after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6

Field data type	Number of fields
Text field, length 20	14
Text field, length 40	2

A maximum of 8 fields are shown for each page.

User field keys are not defined by default in the system. The system has to be customized respectively in order to be able to support this kind of user fields.

Comment tab

The Comment tab allows additional comments about the resource to be stored.

Main tab “resource attributes”

Additional resource attributes are shown using the user field definitions of the resource family. The "resource attributes" button is used for editing.

Resource list main tab

Shows the resource list for the selected resource. By clicking the "resource list" button you can directly go to the BOM application for editing purposes.

Toolbar

General tab

Insert

Opens the dialog for adding resource lists

Copy

Opens the dialog for copying resource lists.

Edit

Opens the dialog for editing resource lists.

Delete

Deletes one or several resources.

Resource tab

Configuration – resource status

Opens the "resource status" application

File - show file

Shows the file - only available for document resources configured as file-based resource without DNC processing in the resource type and if the corresponding license and function authorizations are available.

Go to - resource list

Opens the "resource list" application. The selected resource is entered as default value for the superior resource.

Go to – required resources

Opens the "required resources" application. The selected resource is entered as default value for the required resource.

Go to – cavity assignment

Opens the "cavity assignment" application. The selected resource is entered as default value.

Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.

Functions – measures

Opens the "measures" application.

Functions – status change

Opens the dialog for changing the resource status.

Functions – release of resource

Opens the dialog for releasing a resource.

Functions – stock transfer

Opens the dialog for transferring/relocating a resource

Workplace tab


Configuration – status assignment

Opens the application "status assignment". The selected resource is taken over.


Configuration – counter configuration

Opens the application "counter configuration". The selected resource is taken over.


Configuration – terminal assignment

Opens the application "terminal assignment". The selected resource is taken over.


Entry – reasons

Opens the application "reasons". The selected resource is taken over.


Entry – Operator positions

Opens the application "operator positions". The selected resource is taken over.


Entry – premium indicator

Opens the application "premium indicator". The selected resource is taken over.


Groups - groups

Opens the application "groups". The group of the selected resource is taken over.


Groups – group assignment

Opens the application "group assignment". The selected resource is taken over.


Other – cycle parameter

Opens the application "cycle parameter". The selected resource is taken over.


Other - workforce requirements of workplaces

Opens the application "workforce requirements of workplaces". The selected resource is taken over.

DNC tab

The tab is only available if a DNC resource is selected. These are resources configured as resources with DNC processing in the resource type.

 **Configuration – resource status**

Opens the "resource status" application

 **Configuration - assignment of DNC family to machine**

Opens the application "assignment of DNC family to machine".

 **File - comparison editor**

Opens the comparison editor for the selected resource or resources. Please see below for further information.

 **File - export**

Exports the file entered for the resource. The target file is input using the file explorer.

 **File - import**

Imports the file entered for the resource. The source file is selected using the file explorer.

 **File - viewer**

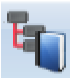
Opens the file entered for the resource for viewing using the defined viewer program.

 **File - editor**

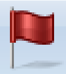
Opens the file entered for the resource for editing using the defined editing program.

 **Go to - resource attributes**

Opens the application "resource attributes". The selected resource is entered as default value.

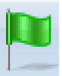
 **Go to - resource list**

Opens the "resource list" application. The selected resource is entered as default value for the superior resource.



Functions – status change

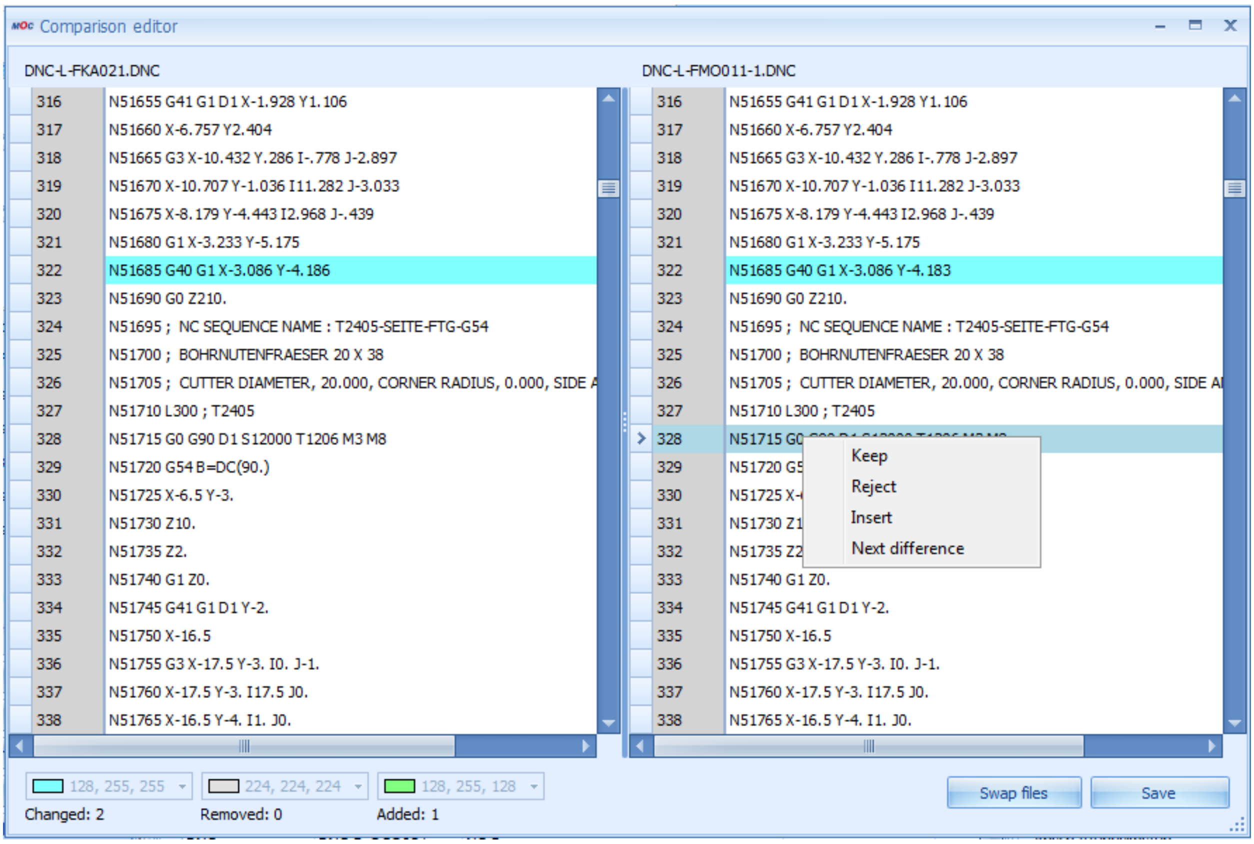
Opens the dialog for changing the resource status.



Functions – release of resource

Opens the dialog for releasing a resource.

Operation of the comparison editor



The comparison editor compares the files attached to the DNC resources. There are two operating modes:

Selection of one resource:

The released resource and the optimized version of the resource are displayed for comparison. The file entered on the right-hand side of the editor can also be changed. Once the changes have been made, they are taken over to the system, just as it is also the case for the simple editor. This mode can only be used for DNC types with the file processing type "optimized".

Selection of two resources:

If two resources are selected, the editor compares the two selected resources. The file type may be selected. The file entered to the right of the editor may also be changed. Once the changes have been made, they are taken over to the system, just as it is also the case for the simple editor.

The functions of the comparison editor can be opened using the relevant buttons or by clicking the right mouse button (context menu):

- Reject: The detected difference (on the right) is rejected. The value from the left file is accepted. The difference is no longer selected.
- Keep: The detected difference (on the right) is accepted. The difference is no longer selected.
- Next difference: goes to the next difference.
- Insert: Inserts a row at the current position.
- Contents of a row can always be changed by clicking the row and inputting values. The row can be left by clicking Esc without making any changes. The row is then highlighted as "changed".
- Swap windows: this button allows for windows to be swapped. This function is necessary if two resources are compared with each other. Their order results from the display order in the table and the system does not know which resource needs to be changed. This button is not available if only one resource is selected. For only the optimized program version can be changed.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration modifications

In order that the settings or modifications made can be interpreted by the terminal shop floor program, the terminal to which the workplace/ machine is assigned must be restarted.

Deleting a machine/ workplace

If data is already recorded and saved for a machine/ workplace, the relevant configuration data of the machine (including the data from the status assignment, the maintenance calendar and the process parameters in the MDE module, among others) is automatically deleted as well.

The previously existing movement data (events, postings) are not automatically deleted with this action; instead, it is deleted later in context of the specified deletion cycles (default: 35 days).

→←

3 Resource Families

Summary

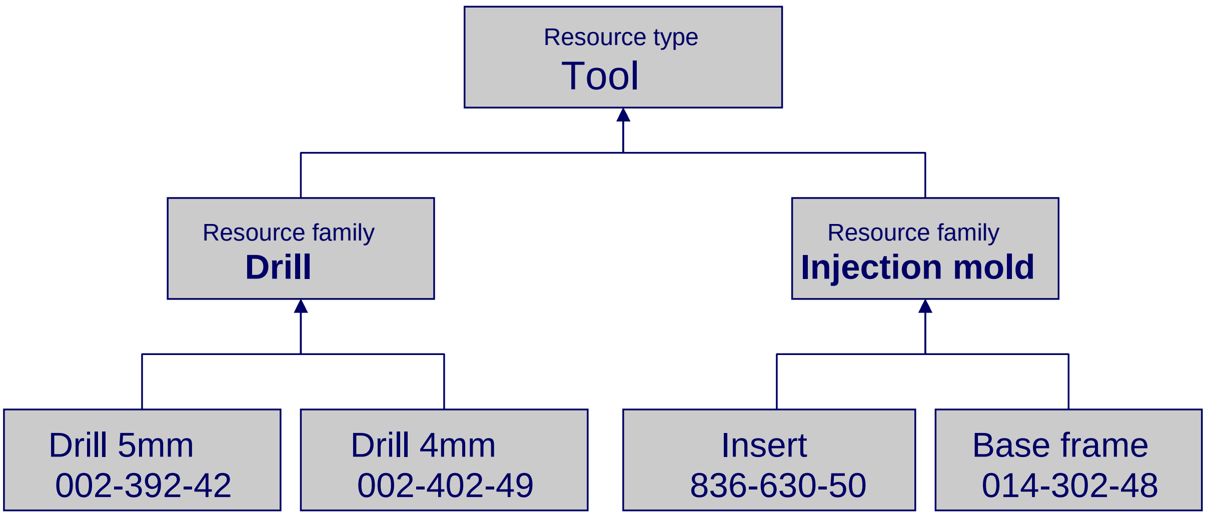
Menu	Master data → Resources → Resource families
Transaction code	resfam.*
Function authorization	mdrfam

This document describes the application "Resource families" within the Manufacturing Operation Center (MOC).

Usage

If the assignment between resource and resource type is assumed, it is clear that in one production company, different resources of the same resource type exist, which also are handled differently. That means that in general classification according to resource type is not sufficient for ordering the resources in a useful way.

By defining so-called resource families, a sub classification of resource types can be created. The following graphic shows an example of how the resource type "Tool" is divided into two resource families, "Drill" and "Injection mold". Each of the individual resources is assigned to one of the two resource families.



Integration

The resource families offer another structural level subordinate to the resource types. The function of the master-detail user fields of the resources that can be defined using the resource types can be refined by definition in the resource families. The resource families are used in particular in the DNC area as a main search criterion and as an assignment criterion on machines.

Selection parameters

In the selection panel, filters can be used for both superior and assigned resources. The following selection criteria are available in the respective application:

Resource type

Type of resource.

Resource family

Family to which the resource is assigned.

Field descriptions

Resource type

Resource type to which the resource families refers

Resource family

Unique "descriptive" designation of the resource family.

This value can be searched for (selected) in the various functions. Because a resource or its resource ID can only be uniquely identified within the resource type, in the evaluations the resource type of the resource is always displayed as well.

Description

Explanation of the resource family; functions as a comment.

Responsibility area

Definition of the responsibility area. By specifying a responsibility area for a family, the responsibility area for the assigned resource is also specified, which controls its visibility and editing options.

Field description for the *General* tab

User field key

Reference to a valid user field key. The specified user field key overwrites the specification in the resource type.

Note regarding DNC filtering using a DNC family and its search fields:

The definition of a suitable user field combination is important for using the flexible filter and search function in the DNC. The definition of such user field combinations is reserved for MPDV customizing. The assignment and use of this user field key is the user's responsibility. With the defined search fields, the DNC records are filtered on the terminal in addition to the DNC family of the machine and can be used as search criteria on the console.

Starting with release DNC 7.2, the following preconfigured user field keys will be delivered:

User field key	Description of the search fields
DNC_K	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited
DNC_K_V	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited 4. Version, mandatory field, can be edited
DNC_K_W	Plastic (tool reference only): 1. Tool, mandatory field, cannot be edited
DNC_K_WV	Plastic (tool reference and version): 1. Tool, mandatory field, cannot be edited 2. Version, mandatory field, can be edited
DNC_NC	NC programs: 1. Article, mandatory field, cannot be edited
DNC_NC_V	NC programs: 1. Article, mandatory field, cannot be edited 2. Version, mandatory field, can be edited
DNC_NC_M	NC programs 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited
DNC_NCMV	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Version, mandatory field, can be edited
DNC_FREI	1. Search field 1, Text20, mandatory field, can be edited

	2. Search field 2, Text20, optional field, can be edited 3. Search field 3, Text20, optional field, can be edited 4. Search field 4, Text20, optional field, can be edited

Note regarding DNC administration

DNC records are used exclusively on machines. To prevent errors in input and assignments, a fixed assignment of a DNC resource family to every machine is carried out and saved in the data of the machine resource (field *Resource family DNC*). In this way, it can be ensured that only programs of one resource family and, indirectly, of one resource type can be loaded on the machine.

Furthermore, for DNC records administration criteria are needed that make selection and evaluation easier, so that identification and processing are also easier, and that enable checks. Because different machine types (e.g. injection molding machine, printer, NC machines) can be considered with the DNC administration, a strict specification of these criteria is not useful. For this reason, there is a resource family for which attributes are stored using the user fields; in turn, these attributes describe and specify the variable parameters.

The attributes are used for identification and can be provided with plausibility functions and assignments in order to create a context for the DNC programs with machines and operations (see the "User fields" chapter).

There are parameters, such as temperature or humidity that affect the behavior of the machine and can therefore influence production. These "environmental factors" can also be collected. To do this, further attributes just have to be defined in the user fields.